

Effect of Transcutaneous Acupoint Electrical Stimulation on Post-Hemorrhoidectomy-Associated Pain, Anxiety, and Heart Rate Variability: A Randomized-Controlled Study

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Abstract

Hemorrhoidectomy is the current best treatment for severe hemorrhoids, but it causes significant postoperative pain and anxiety, which is associated with heart rate variability (HRV). Transcutaneous acupoint electrical stimulation (TAES) was assumed to alleviate pain and anxiety, and modify the autonomic nervous system. This study aimed to examine the effects of TAES intervention on postoperative pain, anxiety, and HRV in patients who received a hemorrhoidectomy. A randomized-controlled trial with

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five repeated measures was conducted. The TAES group ($n = 39$) received four 20-min sessions of electrical stimulation at *chengshan* (BL57) and *erbai* (EX-UE2) after hemorrhoidectomy, whereas the control group ($n = 41$) did not. Data were collected using Visual Analogue Scale (VAS), State Anxiety Inventory (STAI), and HRV physiological signal monitor. TAES resulted in a significant group difference in pain scores, anxiety levels, and some HRV parameters. The findings indicate that TAES can help reduce pain and anxiety associated with hemorrhoidectomy. TAES is a noninvasive, simple, and convenient modality for post-hemorrhoidectomy-associated pain control and anxiety reduction.

Keywords

hemorrhoidectomy, transcutaneous acupoint electrical stimulation (TAES), postoperative pain, postoperative anxiety, heart rate variability (HRV), complementary therapy

Introduction

Hemorrhoidal disease, commonly known as hemorrhoids, is the most common anorectal disorder (Lohsiriwat, 2015). The estimated prevalence of hemorrhoids is as high as 38.93% in Austria (Riss et al., 2012), whereas that in Taiwan is 34%, with all cases receiving surgery (P. C. Lee et al., 2013; Taiwan Ministry of Health and Welfare, 2013). External hemorrhoids originate below the dentate line and are painful because they contain somatic innervation; by contrast, internal hemorrhoids originate above the dentate line and are viscerally innervated, rendering them painless. A patient may have both internal and external hemorrhoids simultaneously (Mounsey, Halladay, & Sadiq, 2011). Hemorrhoidectomy is the most common and effective treatment for third- and fourth-degree internal hemorrhoids (Acheson & Scholefield, 2008); however, it is associated with significant postoperative pain from internal anal sphincter spasms (Perrotti, Dominici, Grossi, Cerutti, & Antropoli, 2010). In addition to pain, patients scheduled to undergo surgery often experience emotional distress such as anxiety. Pain and anxiety are associated with autonomic nervous system activity. Transcutaneous acupoint electrical stimulation (TAES) can alleviate postoperative pain and anxiety, and modify autonomic nervous system activity (Yeh, Chung, Chen, & Chen, 2011).

Conventional treatments, such as those involving nonsteroidal antiinflammatory drugs (NSAIDs) and opioid analgesics, are commonly used to manage postoperative pain in patients after a hemorrhoidectomy. However, these

drugs are associated with unwanted and occasionally severe side effects (Solorio-López et al., 2015). Thus, to reduce postoperative pain and side effects, an integrated approach of pharmacological methods and complementary analgesic modalities is recommended. To alleviate pain, TAES may have analgesic potency and an underlying neurobiological mechanism similar to that of electrical acupuncture (R. Zhang, Lao, Ren, & Berman, 2014), in which an acupoint is stimulated using a needleless electrical impulse. Different frequencies of TAES induce the release of opioid peptides with varied analgesic effects. Low-frequency (LF) stimulation induces analgesia through the release of β -endorphin, enkephalin, and orphanin, whereas high-frequency (HF) stimulation induces analgesia through the release of dynorphin (Lin & Chen, 2008). In particular, TAES can be used to provide intermittent stimuli by alternating LF and HF (called dense-and-dispersed) electrical currents. TAES can quantitatively control stimulation time and intensity as well as prevent pain and topical skin tissue injury from acupuncture. Thus, patients accept this treatment modality easily. TAES is associated with a reduction in analgesic consumption after surgery (Y. C. Chung, Chien, Chen, & Yeh, 2014; Yeh, Chung, Chen, Tsou, & Chen, 2010). In addition, a meta-analysis of 21 randomized-controlled trials (RCTs) revealed that analgesic consumption for postoperative pain is lower in patients who receive TAES than in those who receive placebo (Bjordal, Johnson, & Ljunggreen, 2003). However, few studies have investigated the effect of TAES on post-hemorrhoidectomy pain relief (N. Lee, He, Wang, & Yang, 2008; Yu & Ma, 2011).

Perioperative psychological pressure may reduce the pain threshold, and affect immune function and circulatory system stability (Lautenbacher et al., 2010). Anxiety and postoperative pain are moderately intercorrelated and accompanied by autonomic nervous system activity such as heart rate, which is regulated by the autonomic nervous system and reflects sympathetic and parasympathetic nerve activities. Heart rate variability (HRV) is a quantitative measure of the functional status of the autonomic nervous system (Camm et al., 1996). Numerous studies have demonstrated that acupuncture controls autonomic nervous system function, including HRV and muscle sympathetic nerve activity. Acupuncture can be used to treat autonomic-nerve-related diseases by modulating the imbalance between sympathetic and parasympathetic activity (Ng, So, Gao, Wong, & Ho, 2008). Changes in parasympathetic nervous activity correlate with the amount of *de qi* sensations during acupuncture manipulation (Hori, Takamoto, Urakawa, Ono, & Nishijo, 2010). Acupoint stimulation can relieve pain and increase parasympathetic activity (Arai et al., 2013; Matsubara et al., 2011; Y. J. Wang, Hsu, Yeh, & Lin, 2013; Y. J. Wang, Wang, & Yeh, 2016). However, the degree to which acupoint stimulation affects the autonomic nervous system remains unknown,

probably because some of its effects depend on the *de qi* sensations (Yeh, Chen, & Lin, 2004). In addition, severer postoperative pain and higher anxiety levels can increase the imbalance in the autonomic nervous system (Gorman & Sloan, 2000). HRV measurements may be used to approximate anxiety levels in patients (Ushiyama et al., 2008).

Study Aim and Hypothesis

This study aimed to provide the TAES intervention to patients who received a hemorrhoidectomy, and then to examine its effect on postoperative pain, anxiety, and HRV in them. We hypothesized that after controlling for covariates, TAES intervention would improve pain, anxiety, and HRV parameters in patients with hemorrhoidectomy.

Method

Research Design and Sample

The present study had an RCT design. A concealed allocation procedure was used to assign participants randomly. After surgery, all the participants received the same routine care, including regular analgesics (oral NSAIDs, intravenous NSAIDs, and intramuscular morphine) and a warm sitz bath. Of both study groups, only the TAES group participants received four 20-min sessions of electrical stimulation at two acupoints after their hemorrhoidectomies. To achieve a balanced number of participants in both study groups, a statistician and researcher, both uninvolved in study recruitment or intervention processes, performed the group allocation through permuted-block randomization, with every block comprising four participants. A randomization list drawn through a computer-generated random combination was used to allocate sequences for randomization. The sequences were placed in opaque sealed envelopes to ensure that the participants were randomly assigned to the groups. However, the participants and researchers could not be blinded to this assignment.

All the participants had received a diagnosis of third- or fourth-degree hemorrhoids, underwent lumbar anesthesia, had an American Society of Anesthesiologists (ASA) physical status classification grade of $\leq$ III, were aged $\geq$ 20 years, and had no swelling, infection, fistula, or ulcer in the acupoint regions. Patients who were receiving anticoagulants and those who had severe hypertension, ischemic heart disease, arrhythmia, blood disease, or pacemakers were excluded. Clinically important changes were made using Cohen's guidelines, whereby 0.2, 0.5, and 0.8 denoted a small, medium, and

large effect size, respectively (Cohen, 1988). Sample size was calculated based on the pain intensity as the primary outcome by using G-Power 3.1.2, in which a small effect size of $f = 0.25$ was expected to achieve a statistical power of 0.80 with a statistical significance of .05; the results showed that each group required 34 participants. To account for a loss of 20%, the minimum sample size requirement was determined to be 82 participants. Figure 1 shows a flow diagram of the research design and study participants. At the beginning of the study, 89 participants were assumed to be eligible, but seven of them declined to participate; the remaining 82 signed an informed consent form and were randomly assigned to the TAES group or the control group ($n = 41$ in each group). Two participants in the TAES group were excluded because fistulas were discovered during surgery. Finally, 80 of the participants completed the study (39 in the TAES group and 41 in the control group), with an attrition rate of 2.44%.

Ethical Considerations

The study hospital's human research ethics committee approved this research (Reference No. TCHIRB-1010401), and all participants provided written informed consent after receiving an explanation of the study. No safety concerns were anticipated. Participants were informed that all personal data would remain confidential at all times, and that they were free to withdraw from the study any time without affecting their care.

Intervention

Two acupoints—*chengshan* (BL57) and *erbai* (EX-UE2)—were selected for the TAES interventions. *Chengshan* is most commonly stimulated for treating anal diseases, and can be effective in reducing pain after anorectal surgery, such as hemorrhoidectomy (Yeh et al., 2004; Zheng, 2003). *Erbai* can be effective against hemorrhoids and rectal prolapse (Yeh et al., 2004; Zhu, 2011). Each TAES intervention was applied for 20 min. The first and second TAES interventions were performed the 4th and 6th hr after surgery. A third and fourth intervention was performed at 7 a.m. and 11 a.m. on the day after surgery.

Participants received the TAES interventions in a lying down position. TAES was applied at the acupoint by using a transcutaneous electrical nerve stimulator (D0205KL, Ching-Ming Co., Taiwan) with two pairs of disposable electrode pads (3×3 cm). The dense-and-dispersed stimulation waves were delivered at a 0.06 to 2.30 mA current and 2 or 100 Hz frequency, depending on the tolerance level of each participant (Chen, Liu, & Lin, 2005; Yeh et al., 2010), with a pulse duration of 0.25 ms. The presence of *de qi*

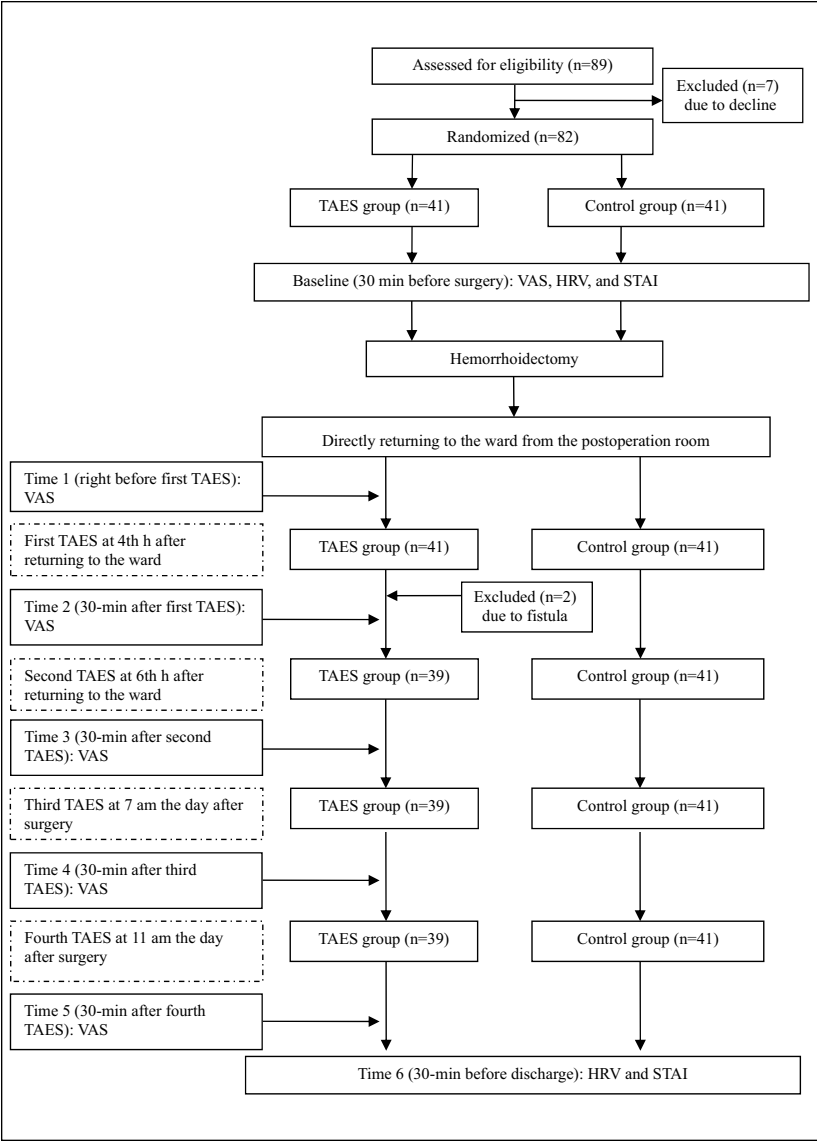


Figure 1. Flow chart of research design, participant recruitment, intervention, and measurements.

Note. TAES = transcutaneous acupoint electrical stimulation; VAS = Visual Analogue Scale; HRV = heart rate variability; STAI = State Anxiety Inventory.

sensations was used to establish the efficacy of the acupoint stimulation. In other words, the participants subjectively reported the sensation of soreness, numbness, swelling, or pain (Yeh et al., 2004). The TAES intervention was performed by a research nurse, who was a qualified member of the research team, and then verified by two traditional Chinese medicine physicians. Moreover, all the participants received routine care (such as dressing, education, etc.) after the hemorrhoidectomy. The control group received no acupoint stimulation.

Outcome Measures

The measured demographic variables and intraoperative data were age, sex, marital status, education level, smoking status, exercise habit, degree of hemorrhoidal disease, pain intensity before surgery, ASA physical status classification score, preoperative hemoglobin level, preoperative hematocrit, anesthesia time, anesthetic dose, and surgical methods. Pain intensity was assessed using the Visual Analogue Scale (VAS), with scores ranging from 0 (*no pain*) to 10 (*very severe pain*). Anxiety levels were measured using the State Anxiety Inventory (STAI; Spielberger, Gorsuch, Lushene, Vagg, & Jacobs, 1983), which reflects the emotional state through the changes in the intensity of environmental stimuli. The 20 items in the STAI are scored on a 4-point Likert-type scale, ranging from 1 (*not at all*) to 4 (*very high*). The overall score ranges from 20 to 80, with scores of 20 to 39, 40 to 59, and 60 to 80 indicating low, moderate, and high anxiety levels, respectively. The STAI has Cronbach's α of .83 to .92 (Spielberger, Gorsuch, & Lushene, 1970); in the present study, Cronbach's α was .88.

HRV was measured for 5 min by using a physiological signal recorder (HRVm, 8Z11, Yang Yin Inc., Taipei, Taiwan), which measures both parasympathetic and sympathetic activities. HRV parameters were as follows: standard deviation of all normal-normal intervals (SDNN), which assesses overall HRV and total activity index of autonomic nervous system function; LF power (0.04-0.15 Hz), associated with the clinical response of sympathetic and parasympathetic activities; HF power (0.15-0.40 Hz), an index of parasympathetic activity; and the LF/HF ratio, which reflects the sympathovagal balance and sympathetic modulations (Camm et al., 1996). Participants were asked to avoid food, alcohol, caffeinated beverage, and nicotine consumption as well as exercise for at least 2 hr before the time of measurement. Short-term HRV was monitored in the sitting position after a 10-min resting period in a temperature-controlled (24°C-26°C) room. The HRV monitor recorded the pulse waveforms for 5 min, while the patients remained silent and motionless with their eyes closed but not asleep. Waveforms were visually inspected to

identify abnormal heart rate or movement artifacts, and then reconfirmed after 15 min when necessary. The HRV parameters were analyzed according to international standards (Camm et al., 1996).

Procedure and Data Analysis

The demographic characteristics, anxiety levels, and HRV parameters were investigated before surgery (baseline). The pain score was measured before the TAES interventions (Time 1) and 30 min after each TAES session (Time 2, Time 3, Time 4, and Time 5). Anxiety levels and HRV parameters were determined again 30 min before discharge from the hospital (Time 6). In the control group, the outcome variables were measured at the same time points as those of the TAES group.

Demographic and intraoperative data were analyzed through a descriptive statistical analysis. The chi-square test was used for categorical comparisons, and the independent *t* test or ANCOVA was used for continuous data. Because postoperative pain scores at repeated measures invalidated the assumption of a normal distribution, a generalized estimating equation (GEE) model was used to analyze the difference in effects of the treatment on the outcomes between the two groups, with controlled covariates and the unstructured correlation matrix into the robust estimates of variance, followed by a Bonferroni test to compare pain scores on each time point. All statistical analyses were performed using IBM SPSS version 20.0 for Windows. A two-sided $p < .05$ was considered statistically significant.

Results

The mean age of the participants was 52.48 ± 15.59 years (range = 23-86 years), and 60% of the participants were women. No significant between-groups differences were noted in the demographic characteristics and intraoperative data ($p > .05$; Tables 1 and 2). At baseline, there were no significant between-groups differences in VAS scores ($p = .97$), anxiety levels ($t = 0.65$, $p = .52$), or HRV parameters, including SDNN ($t = -0.05$, $p = .96$), LF ($t = 0.32$, $p = .75$), HF ($t = 0.23$, $p = .82$), and LF/HF ratio ($t = 0.44$, $p = .67$).

Figure 2 presents the VAS scores measured at different time points in both groups. Moreover, Table 3 shows the result of the GEE model for postoperative pain. After controlling for age, gender, hemorrhoid degree, and surgical treatment, the between-group difference in VAS scores was significant over time ($p = .005$). Trend differences (time-by-group interactions) in the VAS scores were significant at Time 2 ($p = .004$), Time 3 ($p < .001$), and Time 4 ($p < .001$). Followed by individual ANCOVA comparing groups at each time point was

Table 1. Comparisons of Demographic Characteristics Between Two Groups.

Variables	TAES (<i>n</i> = 39)	Control (<i>n</i> = 41)	<i>t</i> or χ^2 (<i>p</i>)
Age (years, <i>M</i> $\pm$ <i>SD</i>)	55.64 $\pm$ 14.83	49.32 $\pm$ 16.34	<i>t</i> = -1.8 (.08)
Sex (<i>n</i> , %)			χ^2 = .05 (.82)
Female	24 (61.5)	24 (58.5)	
Male	15 (38.5)	17 (41.5)	
Marital status (<i>n</i> , %)			χ^2 = .15 (.59)
Married	31 (79.5)	27 (65.9)	
Unmarried	6 (15.4)	12 (29.3)	
Other	2 (5.2)	2 (4.8)	
Education level (<i>n</i> , %)			χ^2 = .24 (.33)
Elementary	2 (5.1)	6 (14.6)	
Junior high	4 (10.3)	6 (14.6)	
Senior high	8 (20.5)	8 (19.5)	
College and above	25 (64.1)	21 (51.3)	
Smoking (<i>n</i> , %)			χ^2 = .04 (.95)
Never	31 (79.5)	32 (78.0)	
Quit	3 (7.7)	3 (7.4)	
Yes	5 (12.8)	6 (14.6)	
Exercise habits (<i>n</i> , %)			χ^2 = .06 (.96)
Not regular	14 (35.9)	13 (31.7)	
<3 times/week	13 (33.3)	15 (36.6)	
3-5 times/week	8 (20.5)	8 (19.5)	
>5 times/week	4 (10.3)	5 (12.2)	

Note. TAES = transcutaneous acupoint electrical stimulation.

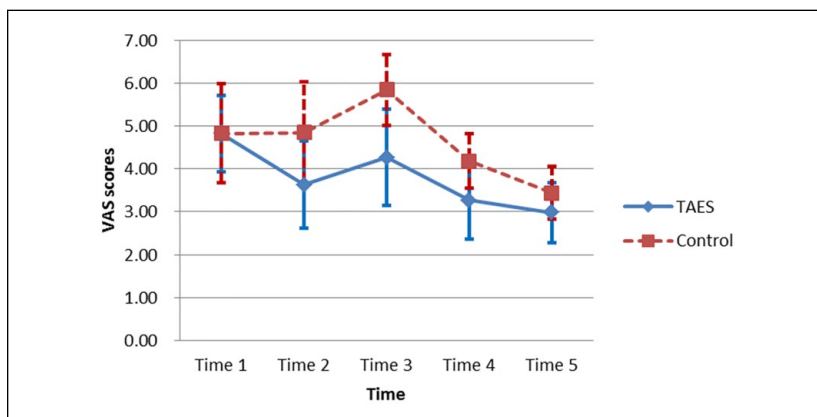
performed, adjusting for Time 1, age, sex, hemorrhoid grade, and surgical treatment, and correcting by a Bonferroni method. As seen in Table 4, the VAS score between two groups differed significantly at Time 2 ($p < .001$), Time 3 ($p < .001$), Time 4 ($p < .001$), and Time 5 ($p = .002$).

In addition to the regularly prescribed oral NSAIDs for postoperative pain, doses of intravenous analgesics were administered at 31.58 ± 6.88 and 37.75 ± 17.26 mg in the TAES and control groups, respectively; no significant between-groups difference was observed ($t = -1.77, p = .08$). Furthermore, intramuscular morphine was administered at 8.89 ± 1.67 and 8.89 ± 1.92 mg in the TAES and control groups, respectively. Table 4 also presents the results of the ANCOVA for postoperative anxiety levels and HRV parameters. After adjusting for baseline, age, sex, hemorrhoid grade, and surgical treatment, between-groups differences were significant in anxiety ($p < .001$), LF ($p = .005$), and HF ($p = .004$) but not in SDNN ($p = .06$) and LF/HF ratio ($p = .89$).

Table 2. Comparisons of Intraoperative Characteristics Between Two Groups.

Variables	TAES (n = 39)	Control (n = 41)	t or χ^2 (p)
Hemorrhoid degree (n, %)			$\chi^2 = .20$ (.21)
Grade III	37 (94.9)	41 (100.0)	
Grade IV	2 (5.1)	0	
Surgical treatment (n, %)			$\chi^2 = .24$ (.20)
Traditional	26 (66.7)	32 (78.0)	
PPH	7 (17.9)	7 (17.1)	
Harmonic scalpel	6 (15.4)	2 (4.9)	
American Society of Anesthesiologists (n, %)			$\chi^2 = .12$ (.57)
I	19 (48.7)	22 (53.7)	
II	20 (51.3)	18 (43.9)	
III	0	1 (2.4)	
Hemoglobin (M $\pm$ SD)	13.55 $\pm$ 1.73	14.0 $\pm$ 1.59	t = -1.25 (.22)
Hematocrit (M $\pm$ SD)	40.92 $\pm$ 4.33	42.12 $\pm$ 3.74	t = -1.33 (.19)
Anesthesia time (min, M $\pm$ SD)	14.36 $\pm$ 2.05	13.9 $\pm$ 2.37	t = 0.92 (.36)
Anesthetic dose (mg, M $\pm$ SD)	9.64 $\pm$ 0.71	9.68 $\pm$ 0.65	t = -0.28 (.78)

Note. TAES = transcutaneous acupoint electrical stimulation; PPH = procedure for prolapse and hemorrhoid.

**Figure 2.** VAS scores in the TAES and control groups.

Note. VAS = Visual Analogue Scale; TAES = transcutaneous acupoint electrical stimulation.

Table 3. The Results of the GEE Model for Pain Scores.

Parameters	β	SE	95% CI
Intercept	4.45	0.32	[3.82, 5.08]
Age	-0.01	0.05	[-0.02, -0.01]
Male vs. female	-0.33	0.14	[-0.61, -0.04]
Hemorrhoid Grade IV vs. Grade III	-0.85	0.22	[-0.128, -0.42]
PPH vs. tradition	0.43	0.24	[-0.03, 0.89]
Harmonic scalpel vs. tradition	0.21	0.13	[0.05, 0.47]
TAES vs. control group	0.42	0.15	[0.13, 0.72]
TAES $\times$ Time 2 ^a	0.74	0.25	[0.24, 1.23]
TAES $\times$ Time 3 ^a	1.08	0.20	[0.70, 1.67]
TAES $\times$ Time 4 ^a	0.45	0.12	[0.22, 0.68]
TAES $\times$ Time 5 ^a	0.34	0.37	[-0.03, 0.11]

Note. Adjusting for baseline, age, sex, hemorrhoid grade, and surgical treatment. GEE = generalized estimating equation; CI = confidence interval; PPH = procedure for prolapse and hemorrhoid; TAES = transcutaneous acupoint electrical stimulation; $\times$ = interaction.

^aReference group: Control group $\times$ Time 1.

Table 4. The Results of ANCOVA in Outcomes.

Variables	TAES (<i>n</i> = 39) <i>M</i> $\pm$ <i>SE</i>	Control (<i>n</i> = 41) <i>M</i> $\pm$ <i>SE</i>	<i>F</i> (<i>p</i>)
Pain at Time 2	3.68 $\pm$ 0.23	4.85 $\pm$ 0.25	76.45 (<.001 ^a)
Pain at Time 3	4.47 $\pm$ 0.38	5.91 $\pm$ 0.41	43.21 (<.001 ^a)
Pain at Time 4	3.49 $\pm$ 0.29	4.34 $\pm$ 0.32	25.02 (<.001 ^a)
Pain at Time 5	2.96 $\pm$ 0.25	3.43 $\pm$ 0.27	10.20 (.002 ^a)
Anxiety at Time 6	40.70 $\pm$ 0.51	40.03 $\pm$ 0.46	23.79 (<.001 ^b)
SDNN at Time 6	29.03 $\pm$ 4.07	24.64 $\pm$ 4.44	3.42 (.06 ^b)
LF at Time 6	246.77 $\pm$ 88.19	98.99 $\pm$ 96.51	8.11 (.005 ^b)
HF at Time 6	144.58 $\pm$ 47.47	63.60 $\pm$ 52.01	8.42 (.004 ^b)
LF/HF ratio at Time 6	2.78 $\pm$ 0.41	2.75 $\pm$ 0.45	0.02 (.89 ^b)

Note. TAES = transcutaneous acupoint electrical stimulation; SDNN = standard deviation of all normal-normal interval; LF = low frequency; HF = high frequency.

^aAdjusting for baseline, age, sex, hemorrhoid grade, and surgical treatment, and correcting by a Bonferroni method.

^bAdjusting for baseline, age, sex, hemorrhoid grade, and surgical treatment.

Discussion

This randomized-controlled study evaluated the effects of TAES on post-operative pain, anxiety, and HRV outcomes in patients who received a

hemorrhoidectomy. The results show that TAES reduces pain for 2 consecutive days after a hemorrhoidectomy, thus indicating that TAES can be effective for pain management. Patients generally stayed in the anesthesia recovery room for approximately 30 min after surgery; however, the pain intensity peaks at 4 to 7 hr postoperatively (Yeh et al., 2010). This study supports that TAES started reducing the post-hemorrhoidectomy-associated pain intensity starting in the first few hours after returning to the inpatient ward; this finding is consistent with previous reports. Transcutaneous electrical nerve stimulation provided effective and long-lasting pain relief for patients after hemorrhoidectomy (Chiu et al., 1999), whereas acupuncture reduced clinical symptoms in patients with hemorrhoids (Zheng, 2003). In addition, acupressure reduced chronic neck pain (Matsubara et al., 2011). The mechanism underlying the analgesic effects of acupuncture has shown that acupoint stimulation prompts the pituitary gland to release opium-derived biological substances such as β -endorphins (Yeh et al., 2010).

In the present study, NSAIDs were administered regularly for pain relief after surgery; nevertheless, additional pain medication was administered if required. However, the participants in the TAES group required slightly fewer additional analgesics (NSAIDs or morphine) than those in the control group did. To the best of our knowledge, no study similar to our study has been published; in addition, our findings are similar to previous reports. Transcutaneous electrical nerve stimulation reduced analgesic consumption after hemorrhoidectomy (Chiu et al., 1999) and total knee arthroplasty (Rakel et al., 2014). Electroacupuncture reduced morphine consumption at 24 hr (Yeh et al., 2011) and 72 hr (Chung, Chien, et al., 2014) after lumbar spine surgery.

As mentioned, anxiety is moderately associated with postoperative pain, and this study indicates that TAES reduced postoperative anxiety. To our knowledge, this is the first study to investigate the effects of TAES on anxiety in patients who have undergone a hemorrhoidectomy. This finding is also consistent with those of other studies that have applied other forms of acupoint stimulation or TAES for other diseases, the results of which showed a reduction in anxiety. TAES reduced postsurgical anxiety in patients with gastroparesis syndrome (C. Y. Zhang, 2010) and those undergoing total knee arthroplasty (Rakel et al., 2014). Acupuncture also reduced anxiety before dental treatment (Karst et al., 2007).

With regard to the HRV, patients who received TAES had more effective functions of the autonomic nervous system than did patients treated by routine care in this study. This finding is similar to those of previous studies. Hu (2011) revealed that acupuncture relieves frozen shoulder pain and increases

parasympathetic activity. Y. J. Wang et al. (2013) reported that auricular acupressure increases parasympathetic activity to maintain homeostasis of autonomic function and alleviate pain in primary dysmenorrhea. However, Matsubara et al. (2011) indicated that acupressure might not improve parasympathetic activity in patients with chronic neck pain. Nevertheless, in this study, the decrease in the parasympathetic activity of the participants in the TAES group was lower than that of the participants in the control group. During pain and anxiety episodes, predominant sympathetic activity results in excessive sympathetic activation or parasympathetic withdrawal, adversely affecting the dynamic flexibility of the autonomic nervous system. Stimulation of an acupoint stimulates the vagus nerve, thus increasing parasympathetic activity, modifying autonomic nervous system activity, and maintaining homeostasis during pain and anxiety episodes. It was noted that the effects of sympathovagal balance and sympathetic modulations were not obvious among the TAES participants in this study. This finding is consistent with systematic reviews on acupuncture that did not evidence the beneficial clinical effect of balancing the autonomic nervous system and modulating the sympathetic nervous system in healthy and nonhealthy people (J. W. Chung, Yan, & Zhang, 2014; N. Lee et al., 2008). In contrast, the sympathovagal balance and sympathetic modulations improved using auricular acupressure in women with menstrual pain and distress (Y. J. Wang et al., 2016) and by using auricular acupuncture in patients after hemicolectomy (Arai et al., 2013). In conclusion, this study shows outcome variable results, but a clear evidence that TAES benefits entire effects on HRV is needed.

Limitations

This study has several limitations. First, the threat to internal validity could not be prevented because of the nonblind study design. Second, performance bias could not be prevented because the interventions and data collection were performed by the same researcher. Third, as both study groups comprised patients from the same hospital, our findings might not be generalizable to other hospitals, despite both groups being highly homogeneous. Thus, future double-blind multicenter RCTs are warranted.

Conclusion and Applications

This randomized-controlled study repeatedly measured the effects of TAES, and revealed that TAES can improve postoperative pain, anxiety, and part of HRV in patients who have undergone a hemorrhoidectomy. The TAES intervention at the two acupoints of *chengshan* (BL57) and

erbai (EX-UE2) demonstrated effective lowered post-hemorrhoidectomy pain and anxiety, and effective improved sympathetic and parasympathetic activities. Although further research is required, the findings of the current study suggest that TAES along with pain medication is effective for pain and emotional management after a hemorrhoidectomy. TAES is a simple, nonpharmacological, noninvasive, and convenient modality for post-hemorrhoidectomy-associated pain control, particularly during the peak of post-hemorrhoidectomy pain, achieving a long-lasting pain relief.

Authors' Note

Lun-Chia Hsu and Shuo-Hui Hung equally contributed to this work as corresponding authors.

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Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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